

Notes: Sammon & Shim (JFE 2026)

Index rebalancing and stock market composition: Do indexes time the market?

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1 Big picture: question, mechanism, and headline results

1.1 What is the paper asking?

A value-weighted total-market index is supposed to “hold the market.” But the market’s *composition* changes over time due to issuance, buybacks, IPOs, and delistings. A value-weighted index therefore must *rebalance* in response to those compositional changes.

The paper asks:

Do existing value-weighted indexes, through their rebalancing rules, implicitly “time” the market? If so, how costly is that for index investors, and can alternative index rules improve performance while still tracking the market?

1.2 Key conceptual mechanism

The key insight is that “holding the market” is equivalent to holding a *constant fraction of each firm’s shares outstanding*. When firms issue shares (typically after good performance / high valuations), a value-weighted index mechanically increases its position *after* issuance. When firms buy back shares (often after underperformance / low valuations), the index mechanically reduces its position *after* buybacks. If issuance/buybacks predict returns, this induces adverse market timing.

1.3 Headline empirical takeaways

- Major value-weighted indexes exhibit economically meaningful deviations from the daily-rebalanced total market, measured by *market tracking error (MTE)* on the order of ~ 75 to 300 bps per year (Table 1).
- The long-short portfolios that represent index funds’ rebalancing trades earn large negative returns: the *total rebalancing portfolio* has mean return about -4.6% per year and a large negative alpha (Table 2).
- Because rebalancing trades are only a fraction of AUM (roughly 10%–15%), the implied *performance drag* for index investors is about 46–69 bps per year.
- A “sleepy” index design (rebalance less frequently and/or delay updating shares outstanding by about 1–2 years) can improve returns by roughly 40–50 bps per year, with modest trading-cost changes but higher MTE (Figs. 1–4).

2 Data objects (what the model needs)

The paper combines:

- **Index-fund holdings** (fund-stock positions) at the quarterly frequency,
- **Stock-level data** (prices, returns, shares outstanding, CRSP identifiers),
- **Index returns** (for S&P, Russell, CRSP index families),

- **Asset-pricing factors** (market, FF-style factors, momentum, reversal, and an issuance factor) for factor-adjusted performance (Table 2).

The conceptual framework below only needs prices, shares outstanding, and AUM/flows; the empirical tests additionally require holdings to infer *how* the index funds rebalance.

3 Conceptual framework: the ownership-ratio representation

3.1 Notation

Let stocks be indexed by $i = 1, \dots, N$ and time by t .

- $P_{i,t}$: price of stock i .
- $\text{ShrOut}_{i,t}$: shares outstanding of stock i .
- $\text{MC}_{i,t} \equiv P_{i,t} \text{ShrOut}_{i,t}$: market cap.
- $\text{TotalM}_t \equiv \sum_{i=1}^N \text{MC}_{i,t}$: total market cap.
- AUM_t : assets under management of a fund that perfectly tracks the index.
- $\text{SharesHeld}_{i,t}$: number of shares of stock i held by the index fund.

3.2 Fund ownership ratio and stock ownership ratio (Eqs. (1)–(2))

Define the **fund ownership ratio**:

$$\Omega_t \equiv \frac{\text{AUM}_t}{\text{TotalM}_t}. \quad (1)$$

Define the **stock ownership ratio**:

$$\Omega_{i,t} \equiv \frac{\text{SharesHeld}_{i,t}}{\text{ShrOut}_{i,t}}. \quad (2)$$

3.3 Derivation: value-weighted market holding $\Leftrightarrow$ constant ownership ratio (Eq. (3))

A value-weighted market index allocates weights

$$w_{i,t} = \frac{\text{MC}_{i,t}}{\text{TotalM}_t}.$$

If the fund tracks the index, dollars invested in stock i equal $\text{AUM}_t w_{i,t}$, hence

$$\text{SharesHeld}_{i,t} = \frac{\text{AUM}_t w_{i,t}}{P_{i,t}} = \frac{\text{AUM}_t}{P_{i,t}} \cdot \frac{P_{i,t} \text{ShrOut}_{i,t}}{\text{TotalM}_t} = \frac{\text{AUM}_t}{\text{TotalM}_t} \text{ShrOut}_{i,t}.$$

Divide both sides by $\text{ShrOut}_{i,t}$ to get

$$\Omega_{i,t} = \frac{\text{SharesHeld}_{i,t}}{\text{ShrOut}_{i,t}} = \frac{\text{AUM}_t}{\text{TotalM}_t} = \Omega_t, \quad \forall i. \quad (3)$$

Interpretation. Holding the value-weighted market is equivalent to owning the *same fraction of each stock's shares outstanding*. This representation makes it clear that *changes in shares outstanding* necessarily force an index to trade.

4 When does a value-weighted index trade? Scaling vs rebalancing

The ownership-ratio view implies two distinct trade motives.

4.1 (i) Cash-driven scaling

If the fund experiences net inflows/outflows, it scales all positions up/down (approximately) proportionally to restore the fund ownership ratio Ω_t . Similarly, dividends or delisting proceeds can mechanically change fund cash and lead to proportional scaling trades.

4.2 (ii) Composition-driven rebalancing

Even without flows, when a particular firm changes shares outstanding (issuance, repurchase, stock-based comp, etc.), its stock ownership ratio $\Omega_{i,t}$ changes. To restore $\Omega_{i,t} = \Omega_t$ for all i , the index must *rebalance across stocks*.

4.3 Index-eligible shares outstanding and index design

In practice, indexes often use *index-eligible shares outstanding*, updated at discrete times. Two design degrees of freedom follow naturally:

1. **Rebalancing frequency** (monthly, quarterly, annually, etc.).
2. **Shares-outstanding delay / staleness**: how old is the shares-outstanding information used at a rebalance?

These choices govern *how much* trading happens at each rebalance (trade size) and *when* the index responds to issuance/buybacks (market timing).

5 A rebalancing identity from the ownership-ratio framework (Eq. (4))

5.1 Economic objects

Over a rebalancing window t (e.g., from $t - 1$ to t):

- $r_{\text{mkt},t}$: the value-weighted market return.
- flow_t : fund net flow as a fraction of beginning-of-period AUM.
- $\text{issue}_{i,t}$: net issuance of stock i as a fraction of last period shares outstanding (negative for buybacks).

- $\text{issue}_{\text{mkt},t}$: aggregate dollar net issuance (scaled by last period total market cap).
- $\text{adddrop}_{\text{mkt},t}$: net IPO additions minus delistings (scaled by last period total market cap).

5.2 Derivation sketch

Target shares are always $\text{SharesHeld}_{i,t} = \Omega_t \text{ShrOut}_{i,t}$. Therefore,

$$\frac{\text{SharesHeld}_{i,t}}{\text{SharesHeld}_{i,t-1}} = \frac{\Omega_t}{\Omega_{t-1}} \cdot \frac{\text{ShrOut}_{i,t}}{\text{ShrOut}_{i,t-1}}.$$

Using $\Omega_t = \text{AUM}_t / \text{TotalM}_t$,

$$\frac{\Omega_t}{\Omega_{t-1}} = \frac{\text{AUM}_t / \text{TotalM}_t}{\text{AUM}_{t-1} / \text{TotalM}_{t-1}} = \frac{\text{AUM}_t}{\text{AUM}_{t-1}} \cdot \frac{\text{TotalM}_{t-1}}{\text{TotalM}_t}.$$

Now use the paper's decomposition of how AUM and total market cap evolve over the window:

$$\frac{\text{AUM}_t}{\text{AUM}_{t-1}} = 1 + r_{\text{mkt},t} + \text{flow}_t, \quad \frac{\text{TotalM}_t}{\text{TotalM}_{t-1}} = 1 + r_{\text{mkt},t} + \text{issue}_{\text{mkt},t} + \text{adddrop}_{\text{mkt},t},$$

and

$$\frac{\text{ShrOut}_{i,t}}{\text{ShrOut}_{i,t-1}} = 1 + \text{issue}_{i,t}.$$

Substituting yields the rebalancing condition (for stocks that are not IPOs and do not delist in the interval):

$$\frac{\text{SharesHeld}_{i,t}}{\text{SharesHeld}_{i,t-1}} = \frac{(1 + r_{\text{mkt},t} + \text{flow}_t)(1 + \text{issue}_{i,t})}{1 + r_{\text{mkt},t} + \text{issue}_{\text{mkt},t} + \text{adddrop}_{\text{mkt},t}}. \quad (4)$$

5.3 Interpretation of Eq. (4)

Eq. (4) implies:

- If stock i issues more than the aggregate issuance rate, then $1 + \text{issue}_{i,t}$ is high, so the index *buys* i at the rebalance (relative to other stocks).
- If stock i buys back shares (negative issuance), then $1 + \text{issue}_{i,t}$ is low, so the index *sells* i (relative to other stocks).
- Returns $r_{\text{mkt},t}$ affect how much rebalancing is needed (because AUM and TotalM move with returns), even though returns alone do not create rebalancing.

This is the key bridge to return predictability: if issuance/buybacks forecast returns, then index rebalancing inherits that predictability (potentially in the *wrong* direction).

Table 1

Market tracking error and the market information ratio for major indexes.

	Panel A: Market tracking error (Basis points)				
	All data	1990–1999	2000–2009	2010–2019	2020–2023
S&P 500 (1990–2023)	233.9	259.0	281.1	134.8	233.2
S&P total market (1995–2023)	309.0	173.8	218.5	444.4	204.0
Russell 3000 (2005–2023)	144.9		100.0	173.4	106.9
CRSP total market (2011–2023)	74.6			59.9	99.6

	Panel B: Market information ratio				
	All data	1990–1999	2000–2009	2010–2019	2020–2023
S&P 500 (1990–2023)	−0.058	0.045	−0.198	−0.109	0.130
S&P total market (1995–2023)	−0.159	−0.179	0.099	−0.345	0.064
Russell 3000 (2005–2023)	0.009		0.143	−0.011	−0.057
CRSP total market (2011–2023)	−0.326			−0.509	−0.109

6 Do existing indexes track the market? MTE and MIR (Eqs. (5)–(6))

6.1 Market tracking error (MTE)

Define the market tracking error as the volatility of the deviation between index returns and the daily-rebalanced market return:

$$\text{MTE} \equiv \text{sd}(r_{\text{idx},t} - r_{\text{mkt},t}). \quad (5)$$

This is conceptually different from the usual “index-fund tracking error” because it compares the *index itself* to the market, not an index fund to its benchmark.

6.2 Market information ratio (MIR)

Define

$$\text{MIR} \equiv \frac{\mathbb{E}[r_{\text{idx},t} - r_{\text{mkt},t}]}{\text{MTE}}. \quad (6)$$

Interpretation. MIR is “return per unit of deviating from the market,” analogous to an information ratio with the market as benchmark.

6.3 Table 1 interpretation: magnitudes and what they mean

Table 1 reports MTE (Panel A) and MIR (Panel B) for major U.S. index families.

Panel A (MTE).

- MTE is economically large: e.g., the S&P Total Market index has MTE about 309 bps (annualized) over its full sample.
- The CRSP Total Market index has much lower MTE (about 75 bps over 2011–2023), consistent with it being closer to an “investable market” and incorporating composition changes more quickly.

Panel B (MIR). Across indexes and subperiods, MIR is generally small and often negative. This means that *deviations from the market are not systematically rewarded*; instead, index design may generate deviations that lower average returns.

Link to the paper’s mechanism. Large MTE suggests the index deviates meaningfully from the daily-rebalanced market. Eq. (4) implies that these deviations are tightly connected to how the index responds to issuance/buybacks and other compositional changes.

7 Rebalancing portfolios: measuring the market-timing content of index trades (Eq. (7) and Table 2)

7.1 Four components of an index fund’s holdings evolution

Over a quarter, a value-weighted index fund’s holdings can be decomposed into:

1. the **long-only portfolio held at the end of the previous quarter**,
2. an **intensive-margin rebalancing portfolio** (trades among existing constituents),
3. an **extensive-margin additions/deletions portfolio** (trades due to IPO adds and delists),
4. a **flow-driven scaling portfolio** (proportional scaling due to flows and other cash events).

The paper constructs long-short portfolios corresponding to items (2)–(4) to study whether the trades have predictable returns (i.e., implicit market timing).

7.2 Intensive-margin rebalancing portfolio construction (Eq. (7))

Let j index funds and t index quarters. Let $\text{shares}_{i,j,t}$ denote the number of shares of stock i held by fund j at the end of quarter t .

Define the **signed rebalancing trade** for stock i as the deviation of actual holdings from “perfect scaling” of the prior position:

$$\Delta \text{shares}_{i,j,t}^{\text{rebal}} \equiv \text{shares}_{i,j,t} - (1 + \text{flow}_{j,t} - \text{adddrop}_{j,t}) \text{shares}_{i,j,t-1}. \quad (7)$$

- $\text{flow}_{j,t}$ captures the fund’s flow-driven scaling need.
- $\text{adddrop}_{j,t}$ captures the cash demand/supply created by net index additions vs deletions (which also requires scaling of existing positions).

Long and short legs. For each fund-quarter:

- Long leg: stocks with $\Delta \text{shares}_{i,j,t}^{\text{rebal}} > 0$ (the fund increased holdings more than scaling predicts).
- Short leg: stocks with $\Delta \text{shares}_{i,j,t}^{\text{rebal}} < 0$ (the fund decreased holdings more than scaling predicts).

Weights (trade-size weighted). Let $p_{i,t}$ be the end-of-quarter price. The expected dollar trade size is $|\Delta \text{shares}_{i,j,t}^{\text{rebal}}| p_{i,t}$. Weights in each leg are proportional to this expected dollar trade size.

Table 2

Rebalancing portfolio returns.

	Intensive margin rebalancing			Scaling		Extensive margin rebalancing			Total rebalancing			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Market		0.0781*** (0.021)	0.119*** (0.028)		0.0058 (0.010)	−0.0197 (0.012)		0.0568 (0.038)	−0.0226 (0.046)		0.0695*** (0.022)	0.038 (0.027)
Size			−0.161*** (0.046)			0.0603** (0.030)			0.382*** (0.096)			0.131** (0.058)
Value			−0.116*** (0.043)			−0.0171 (0.021)			−0.119 (0.073)			−0.111** (0.043)
Profitability			−0.169*** (0.057)			0.0499* (0.027)			−0.0245 (0.099)			−0.188*** (0.059)
Investment			−0.117* (0.060)			0.0151 (0.024)			−0.415*** (0.098)			−0.192*** (0.056)
Momentum			0.0252 (0.030)			−0.0357** (0.017)			0.171*** (0.051)			0.0552* (0.030)
Reversal			−0.154*** (0.039)			0.0057 (0.020)			0.0769 (0.070)			0.0016 (0.042)
Issuance			−0.146*** (0.052)			0.0066 (0.025)			0.0385 (0.096)			0.0413 (0.061)
Alpha	−4.149*** (0.638)	−4.950*** (0.671)	−3.202*** (0.701)	0.363 (0.300)	0.304 (0.313)	0.295 (0.340)	−5.635*** (1.123)	−6.217*** (1.165)	−4.889*** (1.299)	−4.612*** (0.642)	−5.326*** (0.645)	−3.974*** (0.669)
Observations	2339	2339	2339	2339	2339	2339	2301	2301	2301	2344	2344	2344
R-Squared	0.000	0.008	0.101	0.000	0.000	0.012	0.000	0.001	0.048	0.000	0.006	0.066

Return measurement. Portfolios are constructed using quarter- t trades and evaluated using returns in quarter $t + 1$.

7.3 Placebo: the scaling portfolio

The scaling portfolio applies the same long/short construction to the *predicted* scaling trades based on flows (and add/drop cash needs), rather than the deviation from scaling. Because flows to index funds are plausibly less informative about future returns than issuance decisions, this portfolio serves as a placebo for a pure price-pressure story.

7.4 Extensive margin: additions and deletions

The additions/deletions portfolio goes long newly added stocks and short deleted stocks (weights based on dollar trade size). This captures the extensive-margin part of compositional updating.

7.5 Asset-pricing regression model behind Table 2

For each portfolio $p \in \{\text{rebal}, \text{scaling}, \text{add/drop}, \text{total}\}$, the paper estimates factor regressions of the form:

$$R_{j,t+1}^{(p)} = \alpha^{(p)} + \beta^{(p)'} \mathbf{F}_{t+1} + \varepsilon_{j,t+1}^{(p)}, \quad (1)$$

where $\mathbf{F}_{t+1}$ includes the market factor and (in richer specifications) size, value, profitability, investment, momentum, short-term reversal, and an issuance factor.

7.6 Table 2 interpretation: what the coefficients imply

(A) Intensive-margin rebalancing is strongly negative. The intensive-margin rebalancing portfolio has:

- Mean return $\approx -4.15\%$ per year (column (1)).
- A large negative alpha even after controlling for many factors: $\approx -3.20\%$ per year (column (3)).
- Significant negative loadings on **Value** (HML), **Profitability** (RMW), **Investment** (CMA), and **Issuance**.

Economic reading: the fund tends to buy (relative to scaling) growth, low-profitability, high-investment, high-issuance stocks and sell the opposite. These are precisely the types of stocks associated with lower future returns in the issuance/repurchase literature. Hence, index rebalancing embodies *adverse market timing*.

(B) The scaling portfolio is close to zero (placebo passes). The scaling portfolio’s mean return and alpha are close to zero (columns (4)–(6)). This weakens an explanation based purely on price pressure from mechanical flow-based trading.

(C) Extensive-margin add/drop trades are also negative. The additions/deletions portfolio has very negative mean returns and alphas (columns (7)–(9)), consistent with prior evidence that additions underperform and deletions outperform.

(D) Total rebalancing is the economically relevant object. The total rebalancing portfolio combines intensive and extensive margins and has mean return about -4.61% per year (column (10)) and a factor alpha around -3.97% (column (12)). This is the key summary statistic for investor welfare because it aggregates all rebalancing-driven trades.

7.7 Back-of-the-envelope: translating rebalancing returns into fund-level drag

Let the rebalancing sleeve be a fraction κ of AUM. If the sleeve earns return R^{reb} (in the long–short sense), the implied performance drag on the whole fund is approximately

$$\text{Drag} \approx \kappa \cdot |R^{\text{reb}}|.$$

Using $|R^{\text{reb}}| \approx 4.61\%$ and $\kappa \in [0.10, 0.15]$:

$$\text{Drag} \approx 0.10 \times 4.61\% \text{ to } 0.15 \times 4.61\% \approx 46 \text{ bps to } 69 \text{ bps per year.}$$

This magnitude is large relative to typical index-fund expense ratios.

8 Designing a better index: hypothetical indexes and the return–tracking trade-off (Figs. 1–3)

8.1 Two design parameters

A hypothetical value-weighted total-market index is characterized by:

1. **Rebalancing frequency:** monthly / quarterly / annual / biennial.
2. **Shares-outstanding delay d :** the minimum number of trading days that must elapse before changes in shares outstanding are incorporated into index-eligible shares used at a rebalance.

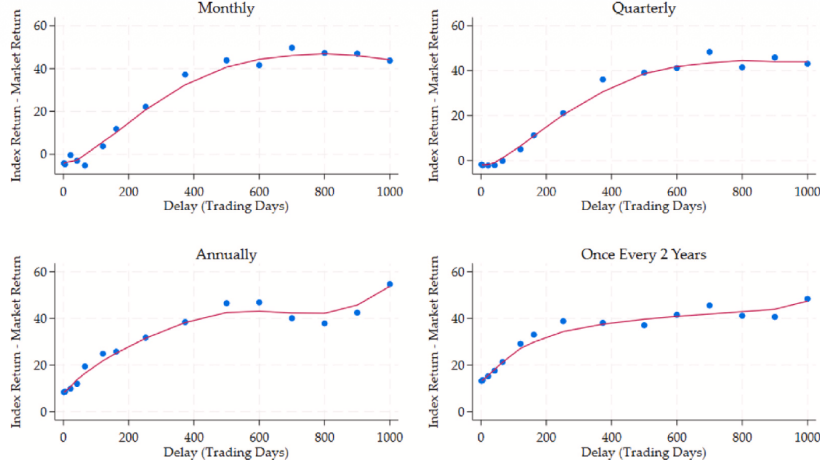


Fig. 1. Excess inde return.

8.2 Performance measures in the simulation

For each pair (frequency, delay), the paper computes:

- **Excess index return** (Fig. 1): index return in excess of the daily total market return, net of expected trading costs.
- **Market tracking error** (Fig. 2): MTE for the hypothetical index relative to the market.
- **Market information ratio** (Fig. 3): $MIR = \text{excess return} / \text{MTE}$ (return per unit tracking error).

8.3 Interpretation of Fig. 1 (excess returns)

Across rebalancing frequencies, longer shares-outstanding delays increase excess returns. The curve is concave: benefits rise quickly up to about a two-year delay (~ 500 trading days) and then plateau. Economically, this means that not immediately “chasing” issuance/buyback activity reduces adverse timing.

8.4 Interpretation of Fig. 2 (tracking error)

MTE rises with longer delays and (mechanically) with less frequent rebalancing. Delays make the index less responsive to compositional changes, increasing deviations from the daily-rebalanced market.

8.5 Interpretation of Fig. 3 (MIR and the optimal interior delay)

MIR typically has an *interior maximum*: small to moderate delays improve returns enough to compensate for higher MTE, but very long delays add tracking error without much additional return. The optimal delay is larger when rebalancing is more frequent, because frequent rebalancing otherwise minimizes staleness.

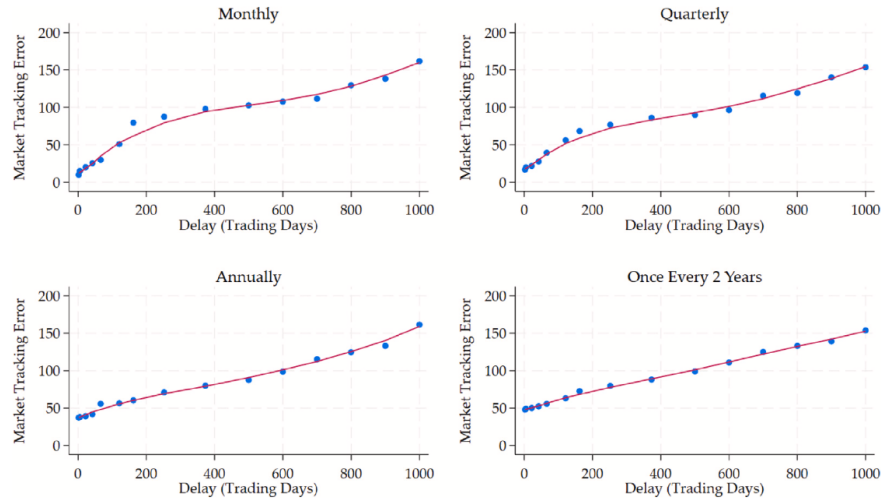


Fig. 2. Market tracking error.

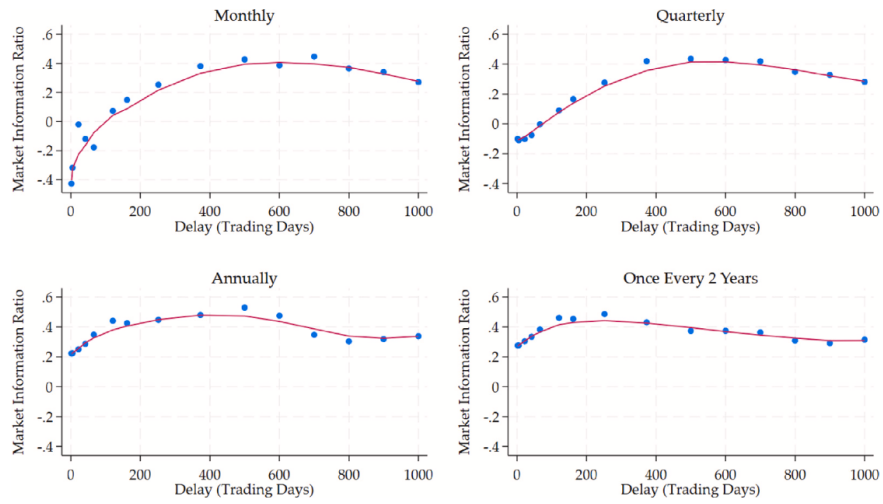


Fig. 3. Market information ratio.

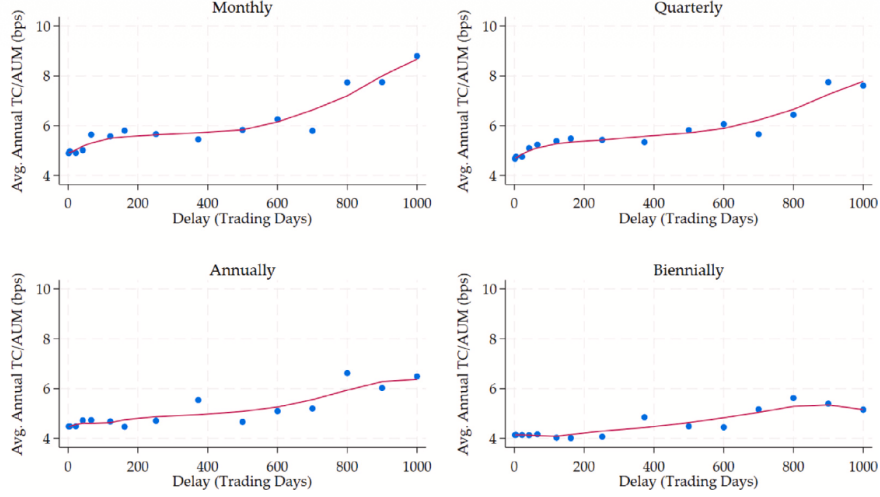


Fig. 4. Estimated trading costs by rebalancing frequency and shares outstanding delay.

9 Trading costs and market timing: a decomposition (Eqs. (8)–(9) and Figs. 4–5)

9.1 Trading-cost model 1: ADV-based price impact (Eq. (8))

The paper estimates expected price impact (percent) for a trade in stock i at time t as

$$PI_{i,t}^{ADV} = \beta \cdot \sigma_{i,t} \left(\frac{\text{TradeSize}_{i,t}}{\text{ADV}_{i,t}} \right)^{1/2}. \quad (8)$$

Here:

- $\sigma_{i,t}$ is the stock's daily return volatility estimated over a short pre-event window,
- $\text{ADV}_{i,t}$ is average daily dollar volume over the same window,
- β is a calibrated coefficient (the paper uses $\beta = 0.2$).

Multiplying $PI_{i,t}^{ADV}$ by the dollar trade size yields a dollar cost; aggregating across trades produces annual trading costs as a fraction of AUM.

9.2 Trading-cost model 2: square-root model with Lambda (Eq. (9))

As an alternative:

$$PI_{i,t}^{\sqrt{}} = \lambda_{i,t} \sqrt{\text{DollarVolume}_{i,t}}, \quad (9)$$

where $\lambda_{i,t}$ is a stock-time liquidity parameter (Lambda).

9.3 Interpretation of Fig. 4 (estimated trading costs)

Trading costs are economically *modest* in AUM terms (roughly single-digit bps per year in the figure). Two robust patterns:

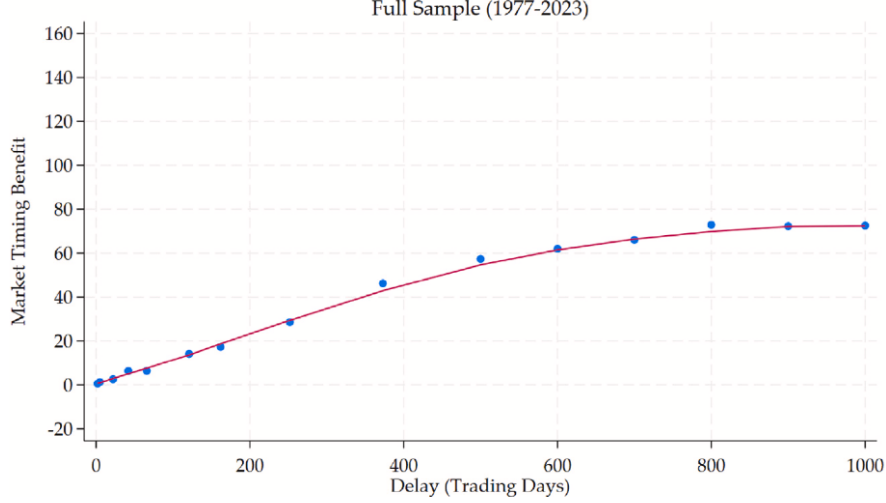


Fig. 5. Market timing.

- **Less frequent rebalancing** tends to *lower* trading costs because price impact is concave in trade size, so fewer, larger trades can be cheaper per dollar traded.
- **Longer delays** can increase trading costs because compositional changes accumulate, requiring larger catch-up trades when the index eventually updates.

A key message is that trading-cost differences are smaller than the return differences in Fig. 1, suggesting market timing dominates.

9.4 Market-timing benefit and Fig. 5

To isolate market timing from trading costs, the paper defines a market-timing benefit for hypothetical indexes that rebalance daily but vary the delay in responding to shares-outstanding changes:

$$\text{MarketTimingBenefit}(d) \equiv \mathbb{E} \left[r^{\text{idx}}(d) - r_{\text{mkt}} \right].$$

Fig. 5 shows this benefit increases with delay and plateaus around a 1–2 year delay. Intuitively, delaying response means the index does *less* of the “buy after issuance / sell after buyback” behavior that is associated with low future returns.

10 Replication via factor timing vs index design

The paper also asks whether delayed rebalancing can be replicated by dynamically trading a market factor and an issuance factor. Even if a dynamic factor strategy can improve Sharpe ratios, it tends to:

- require more deviation from the market (higher tracking error),
- deliver lower average returns without leverage, and

- be less aligned with a “simple passive index” objective.

This supports the practical relevance of *index design* (frequency and shares-outstanding delay) as a mechanism for improving investor outcomes.

11 Summary

1. **Ownership ratio identity:** market-cap weighting $\Leftrightarrow$ constant fraction of shares outstanding.
2. **Key rebalancing equation (4):** index buys high-issuance stocks and sells buyback stocks to restore ownership ratios.
3. **MTE vs MIR:** MTE measures deviation magnitude; MIR measures return per unit deviation.
4. **Empirical punchline:** rebalancing portfolios earn large negative returns ($\sim -4.6\%$), implying a fund-level drag of 46–69 bps.
5. **Design insight:** delaying shares-outstanding updates by 1–2 years and rebalancing less frequently can raise returns by ~ 40 –50 bps, with modest trading-cost changes but higher MTE.